Model Context Protocol (MCP)

A Professional Research Briefing: Architecture, Ecosystem, Governance, and Security

---

Prepared: August 13, 2026

Sources: Model Context Protocol official blog and documentation, The Register, The New Stack, Cloud Security Alliance, MintMCP, Toloka, The Agentics, langprotect.com

Executive Summary

The Model Context Protocol (MCP) is an open-source standard, originally released by Anthropic in November 2024, that lets AI applications connect to external data sources, tools, and workflows through a consistent interface. Often described as a 'USB-C port for AI applications,' MCP has become the de facto standard for AI-to-tool integration, achieving native support across Claude, ChatGPT, Gemini, Copilot, and Cursor. In December 2025 the protocol was donated to the Linux Foundation's Agentic AI Foundation, with OpenAI, Google, Microsoft, AWS, Salesforce, and Snowflake as backers, formalizing it as vendor-neutral infrastructure. The July 2026 specification release (2026-07-28) represents the largest overhaul since launch, moving the protocol to a stateless architecture. Enterprise adoption intent is outpacing production deployment, and security governance has not kept pace with growth — a gap that is now colliding with new regulatory enforcement under the EU AI Act.

1. What MCP Is

MCP is an open-source standard for connecting AI applications to external systems — data sources such as local files and databases, tools such as search engines and calculators, and workflows such as specialized prompts — enabling AI applications to access information and perform tasks through a single, consistent interface rather than bespoke integrations per tool.

The protocol is built on JSON-RPC 2.0 and standardizes how AI agents, acting as clients, connect to and interact with external tools and services, acting as servers.

2. Timeline and Governance

Date	Milestone
Nov 2024	Anthropic open-sources MCP as an internal protocol for connecting AI assistants to external tools.
Nov 2025	Specification version 2025-11-25 formalizes OAuth 2.1 as the authentication standard for remote MCP servers.
Dec 2025	MCP donated to the Linux Foundation's Agentic AI Foundation; OpenAI, Google, Microsoft, AWS, Salesforce, an
Jan–Feb 2026	30+ CVEs filed against MCP servers, clients, and infrastructure components, including a 9.6 CVSS finding in a w
Mar 2026	2026 roadmap published: transport scalability, agent communication, governance maturation, and enterprise rea
Jul 28, 2026	Specification 2026-07-28 finalized: stateless protocol core, header-based routing, formal extensions framework, u
Aug 2, 2026	EU AI Act enforcement begins, with penalties up to 7% of global revenue for AI governance failures.

Protocol development now runs through Working Groups and a formal Spec Enhancement Proposal (SEP) process, reflecting the project's shift from a small maintainer-led effort to community-driven governance.

3. The 2026-07-28 Specification: Key Changes

Described by lead maintainer David Soria Parra as the most substantial revision since the protocol's launch — comparable in scope only to the original addition of authorization — this release delivers on the 2026 roadmap.

- **Stateless core.** MCP is now stateless at the protocol layer, achieved through six coordinated Specification Enhancement Proposals (SEPs), allowing it to run on ordinary HTTP infrastructure instead of requiring persistent sessions.
- **Header-based routing.** Streamable HTTP now requires Mcp-Method and Mcp-Name headers so load balancers, gateways, and rate-limiters can route on the operation without inspecting the request body; servers reject requests where headers and body disagree.
- **Formal extensions framework.** Capabilities such as MCP Apps (server-rendered UI) and Tasks (long-running operations) now ship as extensions on independent timelines rather than being bundled into the core spec. Tasks was moved entirely out of the core protocol.
- **Backward-compatibility risk.** Deprecated features remain functional for at least 12 months, but this is not a blanket interoperability guarantee — servers built on 2026-07-28 may not work with older clients and vice versa.
- **SDK scale.** Tier 1 SDKs are seeing close to half a billion downloads per month, with both the TypeScript and Python SDKs having each crossed one billion total downloads.

4. Enterprise Adoption: Intent vs. Production

Metric	Figure
Enterprise apps projected to include task-specific AI agents by end of 2026 (Gartner)	40%
API gateway vendors projected to have MCP features by end of 2026 (Gartner)	75%
Enterprise apps shipped in Q1 2026 embedding at least one AI agent (up from 33% in 2024)	80%
Organizations with AI agents fully deployed to production	17%
Adoption-intent-to-production gap	~68 pts
Software-industry technical leaders reporting limited-to-broad MCP production use	~41%
Most-searched MCP servers offered as remote, vendor-hosted (vs. local install)	~80%

Server counts and SDK download figures should be treated as adoption-*intent* signals rather than proof of active production usage; organizations are advised to maintain their own internal inventory rather than relying on aggregate registry figures.

5. Security: The Central Risk

Security governance has not kept pace with MCP's adoption curve. Between January and February 2026 alone, researchers filed more than 30 CVEs against MCP servers, clients, and infrastructure components, including one scoring 9.6 on the CVSS scale that affected a widely used proxy package across numerous deployments. This is described by researchers not as an anomaly but as the predictable consequence of deploying a complex, permission-rich protocol at scale before security governance frameworks matured.

Independent scans have found that a majority of public MCP servers carry exploitable risk, with only a small fraction using OAuth authentication by default. The deployment landscape now spans sophisticated enterprise implementations with strong controls alongside thousands of internet-exposed servers with no authentication whatsoever.

Common Enterprise Mitigation Patterns

- Per-agent identity with scoped, independently rotatable credentials, enabling audit attribution separate from human user accounts.
- Shadow AI detection to surface off-gateway MCP usage occurring in developer tools such as Cursor and Claude Code.
- Inline policy enforcement (DLP integration) on every tool call, tying into existing platforms such as Bedrock Guardrails, GCP DLP, Microsoft Purview, Nightfall, and Skyflow.
- Tool-level access control that separates read from write permissions so agents cannot exceed intended scope.
- Centralized MCP gateways providing OAuth/SSO, audit logs, and one-click governed connector deployment.

6. Regulatory Context

EU AI Act enforcement began August 2, 2026, with penalties of up to 7% of global revenue for AI governance failures, making MCP-related security and governance practices effectively mandatory rather than optional for organizations operating in scope of the regulation. This lands against a backdrop where 71% of organizations regularly use generative AI but only 18% have enterprise-wide AI governance councils in place — a governance-containment gap that leaves many companies able to monitor agent activity but unable to block risky operations in real time.

7. Outlook

The 2026 roadmap centers on four priority areas: transport scalability, agent-to-agent communication, governance maturation (including a formal contributor ladder and delegation model), and enterprise readiness. Analysts frame 2026 as the year MCP moves from pilot projects to enterprise-wide production adoption, but caution that the data on how many pilots actually complete that transition is far less optimistic than download and server-count figures alone suggest. The organizations closing the adoption-to-production gap fastest are reported to share one trait: treating evaluation, governance, and human oversight as infrastructure requirements built in from the start, rather than as afterthoughts layered on after deployment.

Sources: modelcontextprotocol.io / blog.modelcontextprotocol.io; The Register (theregister.com); The New Stack (thenewstack.io); Cloud Security Alliance Labs (labs.cloudsecurityalliance.org); MintMCP (mintmcp.com); Toloka (toloka.ai); The Agentics (theagentics.co); langprotect.com. Compiled from web sources as of August 13, 2026; figures and projections should be independently verified for decision-critical use.